

Problem 1

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Problem 1 Determine whether the following statements are true or false and give either an explanation or a counterexample.

- (a) The function $f(x) = \sqrt{x}$ has a local maximum on the interval $[0, 1]$.

Explanation. False. Since $\sqrt{x}$ is increasing over the entire region $[0, 1]$, the only candidate for a local maximum would be $x = 1$. But by definition, endpoints are never local extrema. So $f(x) = \sqrt{x}$ has no local maximum on the interval $[0, 1]$.

- (b) If $f'(2) = 0$, then $x = 2$ is either a local maximum or local minimum of f .

Explanation. False. Consider $f(x) = (x - 2)^3$. This has derivative $f'(x) = 3(x - 2)^2$, so that $f'(2) = 0$. We can see from the graph of f below, that $x = 2$ is not a local extremum for f .

